Distance-Shell Tomography on Graphs: Integer Trades and Optimal Grid Sensing

Lennart Rudolph

September 6, 2026

Abstract

We study recovery of a nonnegative integer population on a known graph from the complete distance histogram at each labelled sensor. Multiplicities are retained, but target identities and correspondence between sensors are unavailable. For every $n \geq 6$, exactly $n-2$ sensors suffice to distinguish all populations of mass at most two on the $n \times n$ Cartesian grid. We prove that an alternating placement on opposite boundaries has integer observation kernel isomorphic to $\mathbb{Z}^4$, with a coordinate-extraction inverse. Its first ambiguous population has mass $n-3$ for even n and $n-2$ for odd n. The resulting four-variable description of every observation fibre gives an exact decoder and separates fixed rational nullity from a growing integer ambiguity threshold. For general graphs, invisible integer trades characterize bounded recovery, and two sensors admit a distance-level cycle characterization. A full-factor Cartesian-product theorem identifies the lifted kernel as a direct sum of factor kernels and preserves every population recovery bound. The accompanying Lean development proves the general trade and product results and, for every $n \geq 6$, the complete integer grid kernel, sharp recovery threshold, optimal sensor count, and four-parameter description of equal-observation populations.

Keywords: population recovery; metric dimension; discrete tomography; integer kernel; sensor placement; Cartesian grid.

1 Introduction

A collection of distance measurements can identify each individual vertex of a graph without identifying a population of vertices. On the 3×3 Cartesian grid, the populations

$$X = \{(1,0),(1,2)\}, \qquad Y = \{(0,1),(2,1)\}$$

both give the distance multiset $\{1,3\}$ at every corner. Two adjacent corners nevertheless identify each single vertex. The missing information is the correspondence between targets seen by different sensors.

We consider a known finite graph and an unknown nonnegative integer population on its vertices. Each labelled sensor reports the full multiset of distances to that population, retaining multiplicities but no target labels. We call this inverse problem *distance-shell tomography*. A nonzero signed population change is an invisible *trade* when every measured shell has sum zero. If $b(G,S)$ is the smallest positive mass of such a trade, then a sensor set S recovers all populations of mass at most h precisely when $h < b(G,S)$. In contrast, recovery without a mass bound is equivalent to full rational column rank of the shell matrix. These general integer-inverse-problem facts motivate a graph-specific question: how does geometry control the smallest realizable ambiguity?

Main result. For the square grid $G_n = P_n \square P_n$, $n \geq 6$, we construct $n-2$ sensors, alternating between two opposite sides, and prove

$$\tau_2(G_n) = n-2, \qquad \ker_{\mathbb{Z}} H_{S_n} \cong \mathbb{Z}^4, \qquad b(G_n, S_n) = \begin{cases} n-3, & n \text{ even}, \\ n-2, & n \text{ odd}. \end{cases}$$

Here τ_h denotes the minimum number of sensors recovering every population of total mass at most h. The kernel description is integral and complete: every invisible population change is reconstructed from four signed vertex occupancies. A bound on its zero interior rows yields the exact ambiguity threshold; the parameterization gives a four-variable description of every observation fibre. Thus the same construction that is optimal for two targets remains optimal for every $2 \leq h < b(G_n, S_n)$. The threshold belongs to this placement; we do not assert that it maximizes b among all placements of size $n-2$.

The contrast with single-vertex localization is unbounded: ordinary metric dimension is two on every such grid. A second contrast is algebraic: rational nullity stays equal to four, while the first integer ambiguity tends to infinity. The proof identifies the mechanism rather than inferring it from finite ranks. A two-sided boundary-response identity reduces the shell equations to two Laurent quantities; degree restrictions leave four parameters, and at most one interior row of a nonzero trade can vanish. Every other interior row has even coefficient sum and therefore contributes at least two to the total absolute mass. Boundary terms complete the sharp bound.

Related observation models. Metric dimension uses a labelled distance vector to identify one vertex [8, 13]. Multiset dimension forgets the landmark labels instead [12]; Hakanen and Yero establish its equivalence with ID-colorings [6]. Our sensors keep their identities, and the anonymous objects are the targets. These are different losses of information, not parameter choices within a single invariant.

The closest several-object comparison is Laihonen's ℓ-set-metric dimension, in which each sensor reports only its distance to the nearest object [7, 9]. His exact grid results and the antecedent cross obstruction are discussed in Section 7. Full histograms are richer reports: they reveal total population as well as every occupied distance shell. Comparisons by number of sensors therefore do not establish equal-cost or equal-bit-rate sensing advantages.

The observation operation itself has established antecedents. Coordinatewise symbol multiplicities are the output of the B-channel [4]. Radial pushforwards of measures, including distance-distribution inverse problems on metric graphs, are studied by Mémoli and Needham [11]. Their inverse questions concern metric-measure spaces; here the graph and sensor locations are fixed, and the unknown is an integer occupancy function on that graph, recovered exactly rather than up to isometry. Class-sum measurements and switching configurations are classical in discrete tomography [5], as are integer moves between equal-observation configurations [2]. We therefore make no priority claim for histograms, integer kernels, or the general circulation mechanism.

The principal contribution is the sharp alternating-grid construction with its full ambiguity lattice and decoder. Two reusable results support it: the exact distance-level cycle description for two sensors, and a Cartesian-product theorem identifying the entire lifted kernel as a direct sum of factor kernels. The latter preserves the first ambiguity exactly, not only bounded injectivity. Detector sets and the sensor-budget profile provide convenient design formulations; their generic equivalences are elementary and are included to make the model self-contained.

Scope and use. The results supply exact obstructions, benchmark sensor placements, and composition rules for an anonymous inverse problem. They can be used wherever the stipulated distance histograms are available, for example in a topology-constrained multiset-output code. A hop-distance event protocol or mesh-feature descriptor would require a separate implementation and measurement justification. We prove no physical ranging, noise robustness, numerical conditioning, or engineering performance result. The most immediate use of the theory is mathematical: selecting sensors by excluding small trades rather than relying on rank alone.

2 Configurations, shells, and trades

Let $G = (V, E)$ be finite, connected, simple, undirected and unweighted, with $N = |V| \geq 2$. Write $\mathbb{N} = \{0, 1, \ldots\}$. A configuration is $x \in \mathbb{N}^V$, of mass $|x| = \sum_v x_v$. Multiple occupancy, targets at sensors, and the empty configuration are allowed. A nonempty sensor set $S \subseteq V$ records

$$(H_S x)_{s,r} = \sum_{v: d(s,v)=r} x_v, \qquad H_S[(s,r), v] = \mathbf{1}_{d(s,v)=r}. \tag{1}$$

This is exactly the multiplicity of r in sensor s's distance bag. Radii $0, \ldots, N-1$ suffice, since a shortest path has no repeated vertex. Zero padding beyond eccentricity adds no information.

For an integer $h \geq 1$, write $\mathcal{X}_h(V) = \{x \in \mathbb{N}^V : |x| \leq h\}$. A set is *$h$-configuration resolving* if (1) is injective on $\mathcal{X}_h(V)$, and $\tau_h(G)$ is its minimum cardinality. Define $\tau_\infty(G)$ analogously for all finite populations. Each sensor reveals $|x|$, so distinct total masses cannot collide. Exactly-h and at-most-h recovery are equivalent here: pad both sides of a smaller equal-mass collision at one fixed vertex. This padding argument depends on allowing multiplicities.

For $z \in \mathbb{Z}^V$, let $z_v^+ = \max(z_v, 0)$ and $z_v^- = \max(-z_v, 0)$. A nonzero vector with $H_S z = 0$ is a *trade*. Summing the rows of one sensor block gives

$$\sum_v z_v = 0, \qquad |z^+| = |z^-| =: m(z) = \tfrac{1}{2}\|z\|_1.$$

Let $b(G, S)$ be the minimum trade mass, or ∞ if no trade exists. We write

$$\mathcal{L}(G, S) = \ker_{\mathbb{Z}} H_S$$

for the *distance-shell trade lattice*. Thus $2b(G, S)$ is its minimum nonzero ℓ_1 norm. This lattice, rather than rational nullity alone, is the central obstruction object.

Proposition 1. *The set S resolves $\mathcal{X}_h(V)$ if and only if $b(G, S) > h$. It resolves all finite populations if and only if* $\operatorname{rank}_{\mathbb{Q}} H_S = N$.

Proof. An equal-observation pair $x \neq y$ gives the nonzero *integer* difference $z = x - y$, with $z^+ \leq x$ and $z^- \leq y$ coordinatewise. Conversely, z^+, z^- give a collision of mass $m(z)$. Full rational column rank implies injectivity. A nonzero rational kernel vector in the deficient case can be multiplied by a common denominator to give an integer trade. □

If a trade exists, the largest guaranteed recoverable mass is $b(G, S) - 1$, not $b(G, S)$. The rank-deficient grid family in Section 5 will show that fixed nullity does not bound this guarantee across graphs of increasing order.

For a nonzero balanced signed population z, define its detector set

$$D_G(z) = \{s \in V : \text{some distance shell about } s \text{ has nonzero signed sum}\}.$$

Every nonzero z has a detector: if $z_v \neq 0$, the radius-zero shell at sensor v detects it. For fixed h, let

$$\mathcal{D}_h(G) = \{D_G(z) : z \neq 0, \ \sum_v z_v = 0, \ m(z) \leq h\}.$$

Proposition 2 (Detector-hypergraph formulation)**.** *A sensor set S is h-configuration resolving if and only if it meets every member of $\mathcal{D}_h(G)$. Hence $\tau_h(G)$ is exactly the transversal number of the finite hypergraph $\mathcal{D}_h(G)$.*

Proof. The condition $S \cap D_G(z) = \varnothing$ says precisely that z is invisible to every sensor in S. Proposition 1 then identifies such a nonzero trade of mass at most h with a collision in $\mathcal{X}_h(V)$. Finiteness follows because every coordinate of a balanced trade of mass at most h lies in $[-h, h]$. $\square$

This exact hitting-set reformulation is elementary but useful: graph geometry enters through the shape of detector sets, while sensor optimization becomes a transversal problem. Lemma 8 will identify one infinite family of detector sets explicitly.

Consequently $\tau_1(G)$ is ordinary metric dimension, and

$$\dim(G) = \tau_1(G) \leq \tau_2(G) \leq \cdots \leq \tau_\infty(G) \leq N - 1. \tag{2}$$

For the final bound, sense all but one vertex: radius-zero counts give those occupancies and total mass gives the last. Adding sensors preserves all distinctions. The hierarchy eventually stabilizes, since there are finitely many sensor subsets and every rank-deficient one has a finite first trade mass.

2.1 The coding interpretation and repeated distance vectors

Order $S = (s_1, \ldots, s_k)$ and define $c_S(v) = (d(s_1, v), \ldots, d(s_k, v))$. The graph supplies a *vertex-indexed* dictionary: two different vertices must remain different inputs even if their distance vectors coincide. A population produces the coordinatewise B-channel type (symbol multiplicity) of that dictionary [4].

Let $C_S = c_S(V)$ be the ordinary set of distinct distance vectors. Write $\mathcal{M}_h(C_S)$ for the multisets of exactly h members of C_S, with repetitions allowed, and let $\mathcal{B}$ record their coordinatewise types.

Proposition 3 (Faithful B-channel interpretation)**.** *For $h \geq 1$, S is h-configuration resolving if and only if both*

$$c_S : V \longrightarrow C_S \text{ is injective}, \qquad \mathcal{B} : \mathcal{M}_h(C_S) \longrightarrow \text{coordinate types is injective.}$$

For the vertex-indexed dictionary, equality of B-channel outputs is always exactly equality of shell observations, without assuming c_S injective.

Proof. At coordinate i, both outputs count the vertices of each distance $d(s_i, v)$ with their occupancies. If $c_S(u) = c_S(v)$ for $u \neq v$, the single-target populations at u, v collide, and padding gives an exactly-h collision. If c_S is injective, pushforward along c_S is a bijection from mass-h vertex populations to $\mathcal{M}_h(C_S)$, preserving observations. Apply the exactly/at-most equivalence proved above. $\square$

The injectivity clause cannot be omitted. On the path 0–1–2 with sensor 1, the collapsed dictionary is $\{(0), (1)\}$. Its one-coordinate type map distinguishes every codeword multiset, but the original vertices 0 and 2 are indistinguishable. Thus a set of distinct codewords cannot silently replace a vertex-indexed dictionary. With an injective dictionary, restricting to binary populations of fixed cardinality gives the usual distinct-codeword B-MAC question; it does not justify multiplicity padding in the binary model.

2.2 The sensor-capacity profile

The minimum-sensor parameter τ_h answers a demand question: how many sensors are needed to recover populations up to mass h? The same theory has a useful dual supply question: for a fixed budget of k sensors, how large a population can be guaranteed recoverable?

Definition 4 (Sensor-capacity profile)**.** For $1 \le k \le N-1$, define

$$\kappa_k(G) = \max\{b(G,S) : S \subseteq V,\ |S| = k\},$$

where $b(G,S) = \infty$ for a full-column-rank placement. We call $(\kappa_k(G))_{k=1}^{N-1}$ the *sensor-capacity profile* of G, indexed by the first ambiguous mass. A finite value κ_k means the best guaranteed mass is $\kappa_k - 1$.

The maximum exists because the graph is finite. Sensor monotonicity implies $\kappa_k(G) \le \kappa_{k+1}(G)$, and sensing all but one vertex gives $\kappa_{N-1}(G) = \infty$.

Proposition 5 (Duality of sensor number and capacity)**.** *For every $h \ge 1$,*

$$\tau_h(G) = \min\{k : \kappa_k(G) > h\}.$$

Proof. By Proposition 1, a k-sensor set resolves all populations of mass at most h exactly when its first trade mass exceeds h. Such a set exists exactly when $\kappa_k(G) > h$. □

In particular, κ_k is an ambiguity threshold, not the mass guaranteed recoverable. The grid consequence $\kappa_{n-3} = 2$ will mean recovery through mass one and unavoidable ambiguity at mass two. This convention accounts for the strict inequality in the duality formula.

3 Two sensors and observation cycles

For distinct sensors s,t, construct a bipartite multigraph $B_{s,t}$. Left vertices are tagged distance levels from s, and right vertices are separately tagged levels from t. Each original vertex v is an *edge identity* joining $(L, d(s,v))$ to $(R, d(t,v))$. Equal endpoint pairs remain parallel edges. Count two parallel edges as a cycle of length two, and give a forest infinite girth.

Theorem 6. *For two sensors,*

$$b(G,\{s,t\}) = \tfrac{1}{2}\,\mathrm{girth}(B_{s,t}).$$

Equivalently, recovery through mass h holds precisely when $B_{s,t}$ has no cycle of length at most $2h$; unrestricted recovery holds precisely for a forest.

Proof. The shell matrix is the unsigned incidence matrix of $B_{s,t}$. Alternating $+1, -1$ on a cycle of length 2ℓ gives a trade of mass ℓ. Conversely, orient every positive trade edge from left to right and every negative edge from right to left, with flow equal to its absolute value. Every level has equal inflow and outflow. Following used outgoing edges and stopping at the first repeated vertex produces a directed cycle with distinct internal vertices. Edge identities are distinct, including in a length-two cycle. Its ℓ positive edges each contribute at least one to the trade, so $\ell \le m(z)$. □

This is a graph-distance specialization of the classical circulation argument, not a new incidence-matrix theorem. Related distance-level zero-divergence and forest elimination appear in Allikvere's different edge-multiset problem [1, Section 6]. In the forest case a decoder repeatedly reads the weight of a leaf's unique edge from its residual margin, subtracts it at the other endpoint, and deletes the edge. Induction proves recovery and uniqueness. Negative inferred weights or nonzero final isolated margins are rejected. A queue uses linearly many graph operations; count arithmetic depends on bit lengths.

The original graph need not contain a cycle. The tree with edges $01, 02, 13, 15, 34, 56$ has distinct distance pairs at sensors $0, 3$, but $\{1, 6\}$ and $\{2, 4\}$ both give $\{1, 3\}$ at each sensor. Its observation multigraph has a four-cycle.

4 Polynomial responses and product constructions

For a signed configuration define

$$P_s^z(q) = \sum_v z_v q^{d(s,v)} \in \mathbb{Z}[q].$$

Its coefficients are exactly the signed shell observations. The square matrix $K_G(q)_{g,u} = q^{d_G(g,u)}$ is the established exponential distance matrix [3]. Its constant coefficient is the identity. Thus multiplication by K_G is injective on polynomial vectors: at the lowest nonzero coefficient of a vector F, the product $K_G F$ has that same nonzero coefficient vector.

Theorem 7 (Exact kernel of a full-factor lift)**.** *Let G, H be finite connected graphs and $\varnothing \neq T \subseteq V(H)$. For $z \in \mathbb{Z}^{V(G)\times V(H)}$, put $z_u(v) = z(u, v)$. Then*

$$z \in \mathcal{L}(G\square H, V(G) \times T) \quad\Longleftrightarrow\quad z_u \in \mathcal{L}(H, T) \quad \textit{for every } u \in V(G). \tag{3}$$

Thus the lifted integer kernel is the direct sum of $|V(G)|$ factor kernels. Moreover,

$$b(G\square H, V(G) \times T) = b(H, T), \tag{4}$$
$$\operatorname{rank}_{\mathbb{Q}} H_{V(G)\times T} = |V(G)| \operatorname{rank}_{\mathbb{Q}} H_T. \tag{5}$$

These statements include the zero-kernel, infinite-threshold case. Consequently, if T resolves through mass h, so does $V(G) \times T$, and

$$\tau_h(G\square H) \leq \min\{|V(G)|\tau_h(H),\ |V(H)|\tau_h(G)\}.$$

Proof. Cartesian distance is the sum of factor distances: every walk must perform at least the required number of steps in each factor, and concatenating factor geodesics attains the bound. For $t \in T$, set

$$F_{u,t}(q) = \sum_v z(u, v) q^{d_H(t,v)}.$$

The vector of responses at (g, t), indexed by g, is $K_G(q)F_t(q)$. Injectivity of this square polynomial mixing matrix shows that all product responses vanish exactly when all $F_{u,t}$ vanish. Equality of polynomial coefficients is equality of signed shell counts, proving (3) in both directions, over $\mathbb{Z}$ and also over $\mathbb{Q}$.

Each invisible slice is balanced because T is nonempty. The positive mass of z is the sum of the positive masses of its slices. If any slice is nonzero, it therefore contributes at least $b(H, T)$. Conversely, place a minimum factor trade in one slice and zero elsewhere. This attains $b(H, T)$ whenever it is finite; if the factor kernel is zero, so is the lifted kernel. This proves (4). Over $\mathbb{Q}$ the direct sum multiplies nullity by $|V(G)|$, and rank–nullity gives (5). The recovery and sensor-number assertions follow from Proposition 1. $\square$

In particular, a trade mixing several first-factor slices cannot lower the ambiguity threshold for this full-factor placement. This conclusion need not hold for an arbitrary proper subset of $V(G)\times T$.

An endpoint sensor recovers every path population. Hence one full side of an $a\times b$ grid recovers every population and $\tau_\infty(P_a\square P_b)\le \min(a,b)$. This is not a rank calculation on a rectangular polynomial matrix: a one-row path response has rational-function rank one but determines every occupancy from its coefficients. Nor is a polynomial replaced by its evaluation at one real number.

We will use the explicit path identity. For $m\ge 1$ and indices $0,\dots,m$, put $K_{ik}=q^{|i-k|}$ and $D=1-q^2$. The tridiagonal matrix T with endpoint diagonal entries 1, interior diagonal entries $1+q^2$, and adjacent entries $-q$ satisfies $TK=DI$. Indeed its interior (i,k) entry is

$$(1+q^2)q^{|i-k|}-qq^{|i-1-k|}-qq^{|i+1-k|},$$

which is D for $k=i$ and zero on either side; endpoint rows are immediate. Therefore

$$\begin{aligned} DF_0 &= Y_0-qY_1,\\ DF_i &= (1+q^2)Y_i-qY_{i-1}-qY_{i+1}\quad (1\le i<m),\\ DF_m &= Y_m-qY_{m-1},\\ &\quad\text{when } Y=KF. \end{aligned} \tag{6}$$

For boundary-grid decoding, coefficients of each F_i follow from its numerator g by $f_j=g_j+f_{j-2}$. Check nonnegativity, the prescribed degree bound, and resynthesis of *all* observations; truncation alone need not reject an inconsistent tail.

5 Optimal sensing on square grids

Write P_n for the path on $0,\dots,n-1$. On $G_n=P_n\square P_n$, shortest-path distance is Manhattan distance. Two adjacent corners recover a single vertex from $i+j$ and $n-1-i+j$. No one sensor suffices, since every grid vertex has two distinct neighbours. Thus $\dim(G_n)=2$ for $n\ge 2$.

5.1 A local obstruction

Lemma 8. *For $n\ge 3$ and an interior centre (a,b), let*

$$X_{a,b}=\{(a-1,b),(a+1,b)\},\quad Y_{a,b}=\{(a,b-1),(a,b+1)\}.$$

The sensors detecting their difference are exactly row a or column b, with the centre removed. Consequently $\tau_h(G_n)\ge n-2$ for every $h\ge 2$.

Proof. For a sensor (u,v) off both lines, put $d=|u-a|+|v-b|$. Both bags are $\{d-1,d+1\}$. On just one line they are $\{d+1,d+1\}$ and $\{d-1,d+1\}$ in one order. At the centre both are $\{1,1\}$. With fewer than $n-2$ sensors, each coordinate projection omits an interior coordinate. Choose missing a,b; every sensor is outside the corresponding detector cross, giving a binary two-target collision. □

The cross geometry has an antecedent in Laihonen's nearest-object grid obstruction [9, Theorem 3]; here the calculation identifies its exact detector set for complete histograms.

Put $m=n-1$, and for $n\ge 6$ use the alternating placement

$$S_n=\{(i,0):1\le i\le m-1,\ i\text{ odd}\}\ \cup\ \{(i,m):1\le i\le m-1,\ i\text{ even}\}. \tag{7}$$

Parity refers to the zero-based coordinate i.

Theorem 9. *For every $n \geq 6$, the integer kernel of H_{S_n} has the explicit integral four-parameter description below, and*

$$\operatorname{rank}_{\mathbb{Q}} H_{S_n} = n^2 - 4, \qquad b(G_n, S_n) = \begin{cases} n-3, & n \text{ even}, \\ n-2, & n \text{ odd}. \end{cases} \tag{8}$$

In particular $\tau_2(G_n) = n - 2$. More generally $\tau_h(G_n) = n - 2$ for every integer $2 \leq h < b(G_n, S_n)$.

5.2 Reduction to four integer parameters

Let $F_i(q) = \sum_{j=0}^{m} z_{ij} q^j$ for an invisible signed configuration. Temporarily work over $\mathbb{Q}(q)$, retaining the requirement that each F_i is a polynomial in degrees $0, \ldots, m$. Define

$$\begin{aligned} L_i &= \sum_k q^{|i-k|} F_k, \qquad & Q_i &= \sum_k q^{-|i-k|} F_k, \\ U &= \sum_k q^{-k} F_k, \qquad & V &= \sum_k q^k F_k. \end{aligned} \tag{9}$$

If R_i is the actual right-boundary response, then $Q_i(q) = q^m R_i(q^{-1})$. The elementary identity $q^{|i-k|} + q^{-|i-k|} = q^{i-k} + q^{k-i}$ gives

$$L_i + Q_i = q^i U + q^{-i} V, \qquad L_0 = V, \quad L_m = q^m U. \tag{10}$$

Hence invisibility prescribes $L_i = 0$ at odd interior indices and $L_i = q^i U + q^{-i} V$ at even ones.

Set $A = U/D$ and $B = V/D$. Applying (6) gives

$$\begin{aligned} &F_0 = B, & &F_1 = -q^3 A - (q + q^{-1}) B, \\ &F_i = (-1)^i (1 + q^2)(q^i A + q^{-i} B) & &(2 \leq i \leq m-2). \end{aligned} \tag{11}$$

The final rows are as follows. For even m, put $C = q^m A$; then

$$F_{m-1} = -(q + q^{-1}) C - q^{3-m} B, \qquad F_m = C. \tag{12}$$

For odd m,

$$F_{m-1} = q^{m-1} A + (q^{1-m} + q^{3-m}) B, \qquad F_m = -q^{2-m} B. \tag{13}$$

Lemma 10. *All invisible configurations, and only those, are the coefficient arrays of (11)–(13) with*

$$\begin{aligned} &n \text{ odd}: \quad B = -a q^{m-2} - b q^{m-1}, \quad q^m A = c q^{m-2} + d q^{m-1}, \\ &n \text{ even}: \quad B = -d q^{m-1}, \quad A = a q^{-2} + b q^{-1} + c, \end{aligned} \tag{14}$$

where $(a, b, c, d) \in \mathbb{Z}^4$ is unique. The same statement holds over $\mathbb{Q}$. Denote its reconstruction matrix by B_n and the extraction map in (15) by E_n; then $E_n B_n = I_4$.

Proof. Suppose first that n is odd, so $m \geq 6$ is even. Both $B = F_0$ and $C = F_m$ initially have support in $[0, m]$. The degree-$m+1$ coefficients in F_1 and F_{m-1} force $B_m = C_m = 0$: their other shifted terms have degree at most three. Degrees at most -2 force $B_j = C_j = 0$ for $0 \leq j \leq m-5$. In particular $B_0 = C_0 = 0$, so degree -1 in those same two rows forces $B_{m-4} = C_{m-4} = 0$. Finally degree -1 in F_2 and F_{m-2} forces $C_{m-3} = B_{m-3} = 0$. Only degrees $m-2, m-1$ remain, giving (14).

Now let n be even, so $m \geq 5$ is odd. Polynomiality of F_m forces $q^{m-2} \mid B$. Put

$$J = q^3 A = -F_1 - (q + q^{-1}) B.$$

This is a polynomial. Equation (13) reads

$$F_{m-1} = q^{m-4}J + (q^{1-m} + q^{3-m})B.$$

Its degree--1 coefficient forces $B_{m-2} = 0$. The second term has degree at most three, so the upper degree bound forces $\deg J \le 4$. Degree $m+1 > 4$ in F_1 now forces $B_m = 0$. Hence B is supported only at $m-1$. Degree -1 in $F_2 = (1+q^2)(q^{-1}J + q^{-2}B)$ forces $J_0 = 0$. Degree $m+1$ in $F_{m-2} = -(1+q^2)(q^{m-5}J + q^{2-m}B)$ forces $J_4 = 0$. Thus $J = aq + bq^2 + cq^3$, as required. All these arguments apply when $n = 6$.

For sufficiency, choose the four parameters and define a proposed L by

$$L_0 = DB, \quad L_m = Dq^m A, \quad L_i = \begin{cases} 0, & i \text{ odd}, \\ D(q^i A + q^{-i}B), & i \text{ even} \end{cases} \quad (1 \le i < m).$$

Set $F = K^{-1}L$. Substitution into the displayed row formulas shows that all exponents lie in $[0, m]$. Moreover $KF = L$, so the *actual* global quantities of these rows are $V = L_0 = DB$ and $U = q^{-m}L_m = DA$. Identity (10) makes every selected even right response zero, while selected odd left responses are zero by construction. Thus these are actual shell-kernel vectors.

The integer parameters are extracted from the coefficient array by

$$\begin{cases} (-z_{0,m-2}, -z_{0,m-1}, z_{m,m-2}, z_{m,m-1}), & n \text{ odd}, \\ (z_{2,0}, z_{2,1}, z_{m-1,m-1}, -z_{0,m-1}), & n \text{ even}. \end{cases} \tag{15}$$

Direct substitution recovers the parameters, including at $n = 6$. This proves integral completeness and independence, not merely rational spanning by cleared columns. The proof over $\mathbb{Q}$ also yields nullity four. □

5.3 Sharp threshold from row support

The row formulas already determine the minimum trade mass without a coefficient census. This gives a short structural reason for the growing recovery threshold.

Lemma 11. *Every nonzero integer trade for S_n has mass at least $n-2$ for odd n and $n-3$ for even n. These bounds are attained by the parameters $(1,0,0,0)$ and $(0,1,0,0)$, respectively.*

Proof. There are $m-3$ interior rows indexed by $2 \le i \le m-2$. By (11), such a row vanishes precisely when

$$q^{2i}A + B = 0.$$

For a nonzero trade, $(A, B) \ne (0, 0)$. If two distinct indices i, j gave zero rows, subtraction would give $(q^{2i} - q^{2j})A = 0$, hence $A = B = 0$, a contradiction. Thus at most one interior row vanishes; none vanishes when exactly one of A, B is zero. Moreover $F_i(1)$ is even, since $1 + q^2$ divides the row in $\mathbb{Z}[q, q^{-1}]$. The coefficient ℓ_1 norm has the same parity as the coefficient sum. Each nonzero interior row consequently contributes at least two to $\|z\|_1$.

Put $\ell = |a| + |b| + |c| + |d|$. When n is odd, the four boundary rows have total coefficient norm at least 4ℓ: each parameter appears in four separate boundary entries, as seen in (11) and (12). These entries are distinct also at $m = 6$. If $\ell = 1$, exactly one of A, B is zero, and

$$\|z\|_1 \ge 2(m-3) + 4 = 2(m-1).$$

If $\ell \geq 2$, allowing one zero row gives

$$\|z\|_1 \geq 2(m-4) + 8 \geq 2(m-1).$$

The unit a has boundary norm four and every interior row has norm two, attaining mass $m-1 = n-2$.

When n is even and $d \neq 0$, the four boundary rows contain four separate copies of d. Hence

$$\|z\|_1 \geq 2(m-4) + 4|d| \geq 2(m-2).$$

When $d = 0$, we have $B = 0$, so no interior row vanishes. The two nonzero boundary formulas supply two separate copies of each of a, b, c, and therefore

$$\|z\|_1 \geq 2(m-3) + 2\ell \geq 2(m-2).$$

The unit b has boundary norm two and every interior row has norm two, attaining mass $m-2 = n-3$. All trades are balanced, so their positive mass is half their coefficient norm. □

5.4 Counting coefficient tracks

Let B_n be the $n^2 \times 4$ coefficient matrix defined by the row formulas. Its rows act on $\boldsymbol{c} = (a, b, c, d)^T$. Opposite coefficient rows $r, -r$, each appearing w times, contribute $w|r\boldsymbol{c}|$ to positive mass.

Lemma 12. *Every nonzero coefficient row occurs with its negative. Their frequencies of* each *sign are given below; the listed expressions mean their coefficient vectors. Unlisted rows are zero.*

n odd		*$n = 2k$ even*	
Expression	*Frequency*	*Expression*	*Frequency*
a, b, c, d *separately*	$n-4$	$a+c$	$k-3$
$a-c,\ b-d$ *separately*	2	$a+c-d$	1
		$a,\ c$ *separately*	$k-2$
		$a-d,\ c-d$ *separately*	1
		b	$n-3$
		d	$n-4$

Proof. For odd $n = 2k+1$, substitute (14) into an interior row. The c track has columns $i-2, i$ and the oppositely signed a track has columns $m-2-i, m-i$. The d, b tracks are their one-column shifts. Before crossings, each parameter has $n-2$ entries of each sign, by counting even and odd interior rows and the four boundary rows. The a, c crossings are exactly

$$(k, k-2),\ (k+1, k-1),\ (k-1, k-1),\ (k, k).$$

They are interior for $k \geq 3$; two carry either sign. The b, d crossings are these sites shifted one column right. Opposite column parities prohibit other intersections. Each parameter loses two occurrences of each sign and each difference gains two, proving the odd table.

For $n = 2k$, the forward interior terms are a at $i-2$, $a+c$ at i, c at $i+2$, and b at $i-1, i+1$, with sign $(-1)^i$. Before d crosses them, each sign occurs $k-1, k-2, k-1, n-3$ times, respectively; d occurs $n-1$ times per sign. Its columns $m-1-i, m+1-i$ have the opposite sign and cannot meet b by parity. The six intersections are

$$\begin{array}{r|cc} a-d & (k, k-2) & (k+1, k-1) \\ c-d & (k-2, k) & (k-1, k+1) \\ a+c-d & (k-1, k-1) & (k, k). \end{array}$$

Each pair has opposite signs. At $k = 3$, two sites lie in rows $1, m-1$; the boundary formulas supply exactly the same c and a terms there. Subtracting these six crossings gives the even table also at $n = 6$. □

The census refines the preceding minimum-mass argument to exact norm formulas for every parameter. For odd n,

$$m(B_n\boldsymbol{c}) = (n-4)(|a| + |b| + |c| + |d|) + 2|a-c| + 2|b-d|. \tag{16}$$

For even $n = 2k$, including $k = 3$,

$$\begin{aligned} m(B_n\boldsymbol{c}) = (k-3)|a+c| + |a+c-d| + (k-2)(|a| + |c|) \\ + |a-d| + |c-d| + (n-3)|b| + (n-4)|d|. \end{aligned} \tag{17}$$

These formulas also describe how the mass changes away from the attaining trades.

Completion of Theorem 9. Lemma 10 proves rank $n^2 - 4$ and identifies every integer trade with a unique nonzero integer quadruple. Lemma 11 proves the exact first trade masses, including matching lower bounds for every nonzero parameter. Proposition 1 gives recovery for every $h < b$. The placement has $n-2$ sensors, and Lemma 8 gives the matching lower bound for $h \geq 2$. Since $b \geq 3$ for $n \geq 6$, the two-target equality follows. The attaining unit-parameter trades use only $0, \pm 1$, so ambiguity at b already occurs between binary populations. □

The capacity in (8) belongs to S_n. It does not assert that no other $n-2$-sensor placement has larger capacity, or determine $\tau_\infty(G_n)$.

6 Four-variable fibres and exact decoding

The integral extraction map does more than establish a minimum trade mass. It reduces every observation fibre to a fixed number of integer variables. In this section $H = H_{S_n}$, and B_n, E_n are the maps from Lemma 10.

Corollary 13. *For every $y \in \mathbb{Q}^{\mathrm{rows}(H)}$ in the image of H, there is a unique $x^0 \in \mathbb{Q}^{n^2}$ satisfying*

$$Hx^0 = y, \qquad E_n x^0 = 0. \tag{18}$$

The integral fibre is empty if x^0 is not integral; otherwise it is exactly

$$\{x \in \mathbb{Z}^{n^2} : Hx = y\} = x^0 + B_n\mathbb{Z}^4.$$

Its nonnegative part is determined by a zero-row check and six pairs of scalar inequalities for odd n, at most eight for even n.

Proof. From any rational solution x, define $x^0 = x - B_nE_nx$. This satisfies (18). The difference of any two normalized solutions belongs to $B_n\mathbb{Q}^4$ and is annihilated by E_n, so is zero. An integral solution x makes $x^0 = x - B_nE_nx$ integral. Conversely, if x^0 is integral, Lemma 10 identifies every integral solution as $x^0 + B_n\boldsymbol{c}$ for a unique $\boldsymbol{c} \in \mathbb{Z}^4$.

Let $\mathcal{R}_n$ contain one representative of each nonzero pair of opposite coefficient rows in Lemma 12. For an occurring row r, set

$$\mu_r = \min\{x_v^0 : (B_n)_{v,*} = r\}.$$

At vertices with zero coefficient row require $x_v^0 \geq 0$. At all other vertices, nonnegativity is equivalent to

$$-\mu_r \leq r\boldsymbol{c} \leq \mu_{-r} \qquad (r \in \mathcal{R}_n). \tag{19}$$

The census supplies six row pairs for odd n, eight for even $n \geq 8$, and seven for $n = 6$ because the $a + c$ row has zero frequency. □

For completeness, this gives an exact decoder with a finite rejection procedure. Reject malformed histograms, unequal sensor totals, or a common total $M > h$. Solve (18) over $\mathbb{Q}$, rejecting inconsistency or a nonintegral x^0. Each coordinate of $E_n x$ is a signed occupancy at a specified vertex, by (15). Hence every admissible parameter belongs to the product of four signed copies of $\{0, \dots, M\}$. Enumerate these at most $(h + 1)^4$ possibilities, test the zero rows and (19), and reconstruct $x = x^0 + B_n\boldsymbol{c}$. Report no solution, the unique solution, or at least two solutions as appropriate.

Every retained candidate is integral, nonnegative, and has observation y. Its mass is M, since the rows of any sensor block sum to total mass. Conversely, every admissible population has exactly one enumerated parameter. This proves soundness, completeness, and termination, including on inconsistent data. Below the threshold in Theorem 9, ambiguity cannot occur. Following one exact rational solve, candidate reconstruction uses $O(n^2(h+1)^4)$ arithmetic operations; this enumeration bound is pseudo-polynomial in a binary-encoded h, not a polynomial bit-complexity or numerical-stability guarantee.

7 Consequences and further questions

7.1 The one-sensor threshold transition

Corollary 14. *For $G_n = P_n \square P_n$ and $n \geq 6$,*

$$\kappa_{n-3}(G_n) = 2, \qquad \kappa_{n-2}(G_n) \geq \begin{cases} n-3, & n \text{ even}, \\ n-2, & n \text{ odd}. \end{cases}$$

Consequently the best guaranteed mass at budget $n-3$ is exactly one. At budget $n-2$, the guarantee is at least $n-4$ for even n and $n-3$ for odd n.

Proof. Two adjacent corners resolve single vertices; enlarge them to $n-3$ sensors, which is possible for $n \geq 6$. The resulting first trade mass is at least two. Lemma 8 gives a mass-two collision for every placement of fewer than $n-2$ sensors. The lower bound at budget $n-2$ follows from the alternating placement in Theorem 9. □

Thus adding one sensor at this budget changes the worst-case recovery guarantee from one target to a quantity growing linearly with n. Separately, Theorem 9 gives a family with constant rational nullity four and $b(G_n, S_n) \to \infty$. This is a separation *across increasing graph orders*; it is not a statement that an arbitrary fixed graph has unbounded thresholds or that the ranks $n^2 - 4$ are equal.

7.2 Comparison with nearest-object observations

For a nonempty population, its histogram determines the nearest occupied shell. Hence our observations determine the nearest-object observations, but not conversely. Laihonen proves that the minimum sensor number distinguishing every nonempty vertex subset of size at most ℓ on an $n \times n$

grid is $n+2$ for $\ell = 2$, and n^2 for $3 \le \ell \le n^2$ [9, Theorems 3–4]. For fixed $h \ge 3$ and sufficiently large n, our $n-2$ sensors distinguish all populations of mass at most h, including multiplicities. On the common binary subdomain this is a quadratic-versus-linear sensor-count contrast between two different information contracts.

There are two qualifications. First, a full histogram reveals more than one distance and also reveals population size. Laihonen's all-vertices obstruction uses subsets of different sizes, so it does not establish the same lower bound when exact cardinality is separately supplied to a nearest-object sensor system. Second, we do not compare report length, bandwidth, or acquisition cost. The comparison identifies information lost by nearest-object reduction, not a hardware-efficiency theorem.

7.3 Four next problems

The following questions concern the present model; they are not claims of a certified historical open-problem list.

Problem 15 (Optimal capacity at the grid threshold)**.** Determine $\kappa_{n-2}(P_n \square P_n)$ and classify the placements attaining it. Determine $\tau_\infty(P_n \square P_n)$ and the values of τ_h after the plateau supplied by Theorem 9.

The alternating placement is sensor-optimal for a substantial range of masses, but its rank deficiency does not prove that every placement of the same cardinality is rank deficient. A matching bound on the graph-wide threshold would require a new obstruction applying to all placements, not only to the displayed kernel.

Problem 16 (Compression of full-factor sensing)**.** For which graph families and factor placements T can a proper subset $S \subsetneq V(G) \times T$ retain the threshold $b(H, T)$? Give sharp sensor savings and characterize the additional trades introduced by omitting first-factor copies.

The full-factor case is already answered exactly by Theorem 7; mixed-slice trades do not lower its threshold. The new issue is whether geometric redundancy allows the same recovery guarantee with fewer sensors.

Problem 17 (Graph-specific primitive trades)**.** Describe or bound the Graver bases of distance-shell matrices for trees, grids, or median graphs, as S varies. Which graph parameters control the support and mass of conformally minimal trades, and can these trades be enumerated or represented efficiently?

Here a Graver vector is minimal under sign-compatible coordinatewise domination; this differs from a primitive circuit, which has minimal support. Every integer-kernel vector admits a sign-compatible decomposition into Graver vectors, a general fact [10, Lemma 3.3], not an open question specific to this model. The graph-specific challenge is to determine those vectors and their quantitative bounds. Two-sensor kernels provide a cycle-based starting class.

Problem 18 (Redundancy and observation errors)**.** Determine the minimum number of grid sensors guaranteeing recovery through mass two after any e sensor erasures. For a specified norm and noise model, determine the minimum separation of distinct mass-at-most-h observation vectors and construct placements with sharp correction guarantees.

At least $n-2+e$ sensors are necessary for the erasure problem: deleting e sensors must leave at least $n-2$. In particular, a sensor-optimal two-target placement tolerates no arbitrary single erasure. Achieving the redundancy bound is not proved here. For additive observation errors, disjoint-ball decoding is standard; the mathematical task is to compute graph-specific separations, not to infer stability from an exact inverse.

8 Computational evidence and verification scope

The reference programs construct graph distances by breadth-first search and compare literal labelled bags with independent shell sums. Exact historical certificates cover $6 \le n \le 12$. Formula regressions check shell cancellation for $6 \le n \le 20$, modular rank through $n = 16$, and the row census through $n = 128$. Separate tests check integral extraction, affine-fibre decoding, duplicate dictionary columns, and the product kernel on small graphs. These are finite implementation checks; the all-order results follow from the preceding proofs.

The companion Lean project uses Lean 4.30.0 and Mathlib v4.30.0. Its public statements are presented separately in `Challenge.lean`; `Solution.lean` proves those same statements from the source development. The verified scope is:

- the literal-bag/shell correspondence and the integer-trade criterion for bounded recovery;
- the complete full-factor product-kernel theorem and equivalence of every mass-bound recovery property, for actual connected simple graphs;
- the grid shortest-path/Manhattan bridge and the $n - 2$ lower bound for every sensor placement and every grid order;
- for every $n \ge 6$, the complete integer kernel with unique four-parameter coordinates and its coordinate-extraction inverse;
- for every $n \ge 6$ and every mass bound, the sharp threshold and the matching optimal sensor count throughout $2 \le h < b(G_n, S_n)$;
- for every $n \ge 6$ and every pair of nonnegative integer populations, equality of observations if and only if their difference has unique four-parameter coordinates.

The universal grid proof works in integer Laurent polynomials. It verifies the local tridiagonal inverse, the equivalence between actual shell sums and alternating row responses, the support restrictions, and both directions of the four-parameter reconstruction. Its minimum-mass proof follows Lemma 11, including the at-most-one-zero-row argument and explicit attaining trades. The general product proof uses polynomial responses and the nonzero constant coefficient of the distance-mixing determinant.

All eight public statements undergo declaration comparison, Lean kernel replay, and independent NanoDa replay. Their transitive axiom reports contain only `propext`, `Quot.sound`, and `Classical.choice`; no proof uses `sorry` or a custom axiom. The standalone challenge contains the expected specification placeholders. The rational-rank clauses, the coefficient census, the two-sensor cycle theorem, the rational normalization in Corollary 13, and executable decoder termination and complexity are written results rather than separately selected Lean theorems. The formal fibre theorem establishes the exact population-level parameterization on which the decoder rests; the reference implementation is checked separately.

Data and software availability. The paper source, Lean proofs, exact reference programs, certificates and validation records are prepared for the Zenodo release under DOI `10.5281/zenodo.22404456`. The release README records reproducible commands, dependency revisions, the formalization coverage and file checksums. The DOI is the identifier reserved for this paper and its companion artifacts.

Declaration of AI assistance. OpenAI ChatGPT and Codex assisted mathematical exploration, derivations, drafting, editing, literature checks, software and Lean proof development. The archived executions are preparation-environment checks, not independent human peer review. The human author is responsible for the claims and final submission; no AI system is an author.

References

[1] Jaan Allikvere. The edge multiset dimension of hypercubes, 2026. URL https://arxiv.org/abs/2608.09983v1. Version 1.

[2] Persi Diaconis and Bernd Sturmfels. Algebraic algorithms for sampling from conditional distributions. *Annals of Statistics*, 26(1):363–397, 1998. 10.1214/aos/1030563990.

[3] Zhibin Du and Rundan Xing. The Graham–Hoffman–Hosoya-type theorems for the exponential distance matrix. *Electronic Journal of Linear Algebra*, 39:62–70, 2023. 10.13001/ela.2023.7449.

[4] Arkadii G. D'yachkov, Nikita Polyanskii, Vladislav Yu. Shchukin, and Ilya Vorobyev. Separable codes for the symmetric multiple-access channel. *IEEE Transactions on Information Theory*, 65(6):3738–3750, 2019. 10.1109/TIT.2019.2893234.

[5] Lajos Hajdu and Robert Tijdeman. Algebraic discrete tomography. In Gabor T. Herman and Attila Kuba, editors, *Advances in Discrete Tomography and Its Applications*, pages 55–81. Birkhäuser, 2007. 10.1007/978-0-8176-4543-4_4.

[6] Anni Hakanen and Ismael G. Yero. Complexity and equivalency of multiset dimension and ID-colorings. *Fundamenta Informaticae*, 191(3–4):315–330, 2024. 10.3233/FI-242185. URL https://fi.episciences.org/13918.

[7] Anni Hakanen, Ville Junnila, Tero Laihonen, and María Luz Puertas. On the metric dimensions for sets of vertices. *Discussiones Mathematicae Graph Theory*, 43(1):245–275, 2023. 10.7151/dmgt.2367.

[8] Frank Harary and Robert A. Melter. On the metric dimension of a graph. *Ars Combinatoria*, 2:191–195, 1976. URL https://combinatorialpress.com/ars/vol2/.

[9] Tero Laihonen. The metric dimension for resolving several objects. *Information Processing Letters*, 116(11):694–700, 2016. 10.1016/j.ipl.2016.06.002.

[10] Jon Lee, Shmuel Onn, Lyubov Romanchuk, and Robert Weismantel. The quadratic Graver cone, quadratic integer minimization, and extensions, 2010. URL https://arxiv.org/abs/1006.0773v1. Version 1; Lemma 3.3.

[11] Facundo Mémoli and Tom Needham. Distance distributions and inverse problems for metric measure spaces. *Studies in Applied Mathematics*, 149(4):943–1001, 2022. 10.1111/sapm.12526.

[12] Rinovia Simanjuntak, Presli Siagian, and Tomáš Vetrík. The multiset dimension of graphs, 2017. URL https://arxiv.org/abs/1711.00225v2. Version 2, revised 2019.

[13] Peter J. Slater. Leaves of trees. *Congressus Numerantium*, 14:549–559, 1975.